Project Title:
Supply Chain Risk & Demand Optimization in Retail

Problem Statement:
Retail businesses often lose revenue due to stockouts and inefficient promotions. This project analyzes sales, inventory risk, and customer behavior to identify demand patterns and improve supply chain decisions.

Objectives:

- **Analyze overall business performance**
- **Identify top-performing products**
- **Evaluate stockout impact**
- **Improve inventory decision-making**

Dataset- Walmart dataset from Kaggle

Analysis & Insights:

1. What is the total revenue generated?
   Insights: The amount of total revenue received is 15263601.45. It shows the performance of the firm as a whole and can act as a standard measure to gauge future growth. An increase or decrease in its value is indicative of shifts in demand, pricing, or customer behaviour.

```
select
round( SUM(Revenue),2) as total_revenue_generated
from walmart
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

total_revenue_generated
15263601.45

2. How many total transactions were completed?
   Insights : total 5000 transactions were completed, this helps measure customer activity and demand.

```
select count(distinct transaction_id) AS Total_transactions
from walmart
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_transactions
5000

3. What is the average order value?
   Insights : The average order value (AOV) from any order placed by the customer is ₹3052. The higher the AOV, the more successful the sales are through upselling or buying an expensive product.

```sql
select avg(Revenue) as avg_order_value
from walmart
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

avg_order_value
3052.720289999991

4. Which Product Categories Generate the Most Revenue?
   Insights: The highest amount of money earned from any category comes from the Electronics category which is 7941631.8. This shows a very high demand for the products in the Electronics category and implies that more promotions can earn more revenue for this category.

```sql
select category, round( SUM(Revenue),2) as total_revenue_generated
from walmart
group by category
order by total_revenue_generated DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_revenue_generated
Electronics	7941631.8
Appliances	7321969.65

5. Which products generate the highest revenue?

Insight: TV is the best performing product, earning ₹2049493.86 in sales. The product has high customer demand, and there is potential for increasing inventory levels, adjusting prices, or launching complementary products.

```
select product_name , round(sum(Revenue),2) as total_revenue
from walmart
group by product_name
order by total_revenue desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	total_revenue
TV	2049493.86
Tablet	1996253.02
Fridge	1938012.69
Smartphone	1931310.04
Washing Machine	1897934.02
Camera	1895104.13
Headphones	1846334.45
Laptop	1709159.24

6. What are the top 5 best-selling products by quantity?

Insight: The best-selling products, in terms of quantity sold, are Fridge, Tablet, TV. These products have high demand. However, high quantity sold does not imply high sales; therefore, price consideration needs to be taken into account.

```
select product_name, sum(quantity_sold) as total_sales
from walmart
group by product_name
order by total_sales DESC
limit 5
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	total_sales
Fridge	1967
Tablet	1964
TV	1926
Smartphone	1876
Camera	1873

7. Which store locations generate the highest revenue?

   Insights: The Store 17 in Dallas, TX branch has the highest sales, earning ₹213954.02. The branch has shown good regional performance and can act as a standard for other branches.

```sql
select store_id, store_location, round( SUM(Revenue),2) as total_revenue_generated
from walmart
group by store_id, store_location
order by total_revenue_generated DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_id	store_location	total_revenue_generated
17	Dallas, TX	213954.02
17	Chicago, IL	211660.03
1	Los Angeles, CA	210976.79
11	Chicago, IL	205997.8
5	Chicago, IL	205576.05
5	Dallas, TX	205035.15
14	New York, NY	203837.41
16	Los Angeles, CA	201977.38
1	New York, NY	201755.28
2	Miami, FL	199495.87
20	New York, NY	195221.76
9	Dallas, TX	187649.41
4	New York, NY	184681.69
8	New York, NY	184609.41
11	New York, NY	183295.13
3	Los Angeles, CA	181196.49
10	Miami, FL	180712.63

8. What Are the Monthly Sales Trends?

   Insights: Sales reach their peak in AUG, which implies that demand is influenced by seasons (such as holidays). Companies need to build up their inventories and marketing strategies during peak seasons.

```sql
SELECT MONTH(STR_TO_DATE(transaction_date,'%d-%m-%Y')) AS month,
    ROUND(SUM(Revenue),2) AS total_monthly_sales
FROM walmart
GROUP BY MONTH(STR_TO_DATE(transaction_date,'%d-%m-%Y'))
ORDER BY month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

month	total_monthly_sales
1	1643241.59
2	1505407.62
3	1603925.59
4	1526409.04
5	1683454.99
6	1556131.39
7	1606099.78
8	1708377.85
9	807882.8
10	553933.33
11	546537.09
12	522200.38

9. Which days of the week generate the most revenue?
Insights: Thursday is the day when sales are at their peak, which implies that this is the day when most people shop. This information can be used for optimizing manpower and marketing strategies.
Sales peak on Thursdays, suggesting an opportunity to optimize staffing, inventory allocation, and promotional campaigns on high-demand days.

```
select weekday , Round(sum(revenue),2) as total_revenue
from walmart
group by weekday
order by total_revenue desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

weekday	total_revenue
Thursday	2436640.72
Monday	2433795.67
Saturday	2222858.28
Tuesday	2126494.17
Wednesday	2095912.06
Friday	2009799.42
Sunday	1938101.13

10. How do daily transaction volumes vary over time?
Insights: Daily transaction volumes fluctuate without a consistent trend, indicating variability in customer demand and potential influence of external factors.

```
SELECT STR_TO_DATE(transaction_date, '%d-%m-%Y') AS formatted_date,COUNT(*) AS total_transactions
FROM walmart
GROUP BY formatted_date
ORDER BY formatted_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

formatted_date	total_transactions
2024-01-01	26
2024-01-02	12
2024-01-03	18
2024-01-04	31
2024-01-05	18
2024-01-06	22
2024-01-07	19
2024-01-08	18
2024-01-09	17
2024-01-13	20
2024-01-14	16
2024-01-15	24
2024-01-16	17
2024-01-17	16
2024-01-18	27
2024-01-19	17
2024-01-20	23

11. Do Loyal Customers Spend More Than Regular Customers?

Insights: Platinum customers generate a revenue of ₹4012963.14, while gold customers generate ₹3536863.54. This indicates that loyal customers have a higher lifetime value, implying that their retention is crucial

```
select customer_loyalty_level ,round( SUM(Revenue),2) as total_spend
from walmart
group by customer_loyalty_level
order by total_spend DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_loyalty_level	total_spend
Platinum	4012963.14
Silver	3918576.32
Bronze	3795198.45
Gold	3536863.54

12. Which age group contributes the most revenue?

Insights: The age group of 31-40 generates the maximum revenue ₹2981156.08. They can be marketed to earn more revenues.

```
SELECT
case
when customer_age BETWEEN 18 AND 20 then '18-20'
when customer_age BETWEEN 21 AND 30 then '21-30'
when customer_age BETWEEN 31 AND 40 then '31-40'
when customer_age BETWEEN 41 AND 50 then '41-50'
when customer_age BETWEEN 51 AND 60 then '51-60'
else '61-70' end as age_group,
COUNT(DISTINCT customer_id) AS customers, Round(sum(Revenue),2) as Revenue
FROM walmart
GROUP BY age_group
order by Revenue desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

age_group	customers	Revenue
31-40	883	2981156.08
41-50	894	2965695.73
51-60	889	2936191.44
21-30	902	2832018.97
61-70	913	2807606.84
18-20	274	740932.39

13. How does customer gender influence purchasing behavior?
    Insights: Female customers generate the highest revenue ₹5193570.83 . This indicates that Female customers have distinct buying habits. This can be used for marketing them.

```sql
select customer_gender, Round(sum(Revenue),2) as Revenue
from walmart
group by customer_gender
order by Revenue DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_gender	Revenue
Female	5193570.83
Other	5191936.83
Male	4878093.79

14. Which income group contributes the most revenue?
    Insights: The High Income generates the highest revenue. This implies that purchasing power greatly impacts sales. Expensive products can be sold to this income group.

```sql
SELECT
CASE
WHEN customer_income < 40000 THEN 'Low Income'
WHEN customer_income BETWEEN 40000 AND 80000 THEN 'Middle Income'
ELSE 'High Income' END AS income_group,
ROUND(SUM(revenue),2) AS total_revenue
FROM walmart
GROUP BY income_group
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

income_group	total_revenue
High Income	6101505.41
Middle Income	6032938.2
Low Income	3129157.84

16. Which Payment Method Is Most Frequently Used?

    Insights: The Credit Card option has been the most frequent choice by customers, thus pointing out their preference for this payment method. Discounts or other incentives offered with respect to this payment mode may increase the sale figures even further.

```sql
select payment_method, Count(distinct transaction_id) as total_transactions
from walmart
group by payment_method
order by total_transactions desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

payment_method	total_transactions
Credit Card	1281
Digital Wallet	1263
Cash	1246
Debit Card	1210

17. Do Promotions Actually Increase Sales?

    Insights: Revenue generated during promotions ₹8062411.03 is greater than those during non-promotions ₹7201190.42. The results suggest that the promotions have been effective in boosting revenue. Promotions generate slightly higher revenue than non-promotional sales, indicating moderate effectiveness. However, the increase is not significant, suggesting scope for optimization.

```sql
select promotion_applied, round(sum(Revenue),2) as total_sales, round(avg(Revenue),2) as avg_sales
from walmart
group by promotion_applied
order by total_sales DESC

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

promotion_applied	total_sales	avg_sales
TRUE	8062411.03	3092.6
FALSE	7201190.42	3009.27

18. Which Promotion Type Generates the Highest Demand?

Insights: Promotion types do not show a strong impact on demand, indicating that factors such as product value and customer preference may play a larger role.

```sql
SELECT
    promotion_type,
    COUNT(*) AS promotion_transactions,
    SUM(actual_demand) AS total_demand,
    ROUND(AVG(actual_demand),2) AS avg_demand
FROM walmart
WHERE promotion_applied = 'True'
GROUP BY promotion_type
ORDER BY total_demand DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

promotion_type	promotion_transactions	total_demand	avg_demand
None	1794	538115	299.95
BOGO	424	122337	288.53
Percentage Discount	389	116295	298.96

19. How much revenue is generated during promotions?

Insights: Revenues from promotions are percentage discount and BOGO are 2400750 and 2534988 respectively.

```sql
select promotion_type , Round(sum(Revenue),2) as Revenue
from walmart
where promotion_type in ('BOGO' , 'Percentage Discount')
group by promotion_type
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

promotion_type	Revenue
Percentage Discount	2400750.12
BOGO	2534988.18

20. What is the overall stockout rate?
Insights: Stockouts occur at a rate of 51.86 %. This means there are too few products available. The higher the stockout rate, the less effective the inventory management.

```sql
SELECT
SUM(CASE WHEN stockout_indicator = 'TRUE' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)
AS stockout_rate
FROM walmart;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

stockout_rate
51.86000

21. Which products frequently experience stockouts?
Insights: The product with the highest number of stockouts is product id 776 i.e Smartphone There seems to be very high demand for these products, but not enough supply. More products need to be ordered.

```sql
select product_id, product_name, category, stockout_indicator, count(*) as stockout_count
from walmart
where stockout_indicator = 'TRUE'
group by product_id, product_name, category
order by stockout_count DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	category	stockout_indicator	stockout_count
776	Smartphone	Appliances	TRUE	4
684	Headphones	Electronics	TRUE	3
727	Headphones	Electronics	TRUE	3
131	Fridge	Electronics	TRUE	3
554	Camera	Electronics	TRUE	3
182	Washing Machine	Electronics	TRUE	3
645	Washing Machine	Electronics	TRUE	3
585	Washing Machine	Electronics	TRUE	3
714	Fridge	Electronics	TRUE	3
447	Camera	Electronics	TRUE	3
942	TV	Electronics	TRUE	2
685	Smartphone	Electronics	TRUE	2
259	Smartphone	Electronics	TRUE	2
157	Washing Machine	Electronics	TRUE	2
526	Washing Machine	Electronics	TRUE	2
142	Fridge	Electronics	TRUE	2
502	Washing Machine	Appliances	TRUE	2

22. Which Stores Have the Highest Stockout Rates?
    Insights: The store with the highest stockout rate is store 14 New York, NY.

```
select store_id, store_location, count(*) as stockout_count
from walmart
where stockout_indicator = 'TRUE'
group by store_id, store_location
order by stockout_count DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_id	store_location	stockout_count
14	New York, NY	48
16	Los Angeles, CA	40
13	Dallas, TX	37
5	Chicago, IL	37
20	New York, NY	37
8	New York, NY	36
5	Dallas, TX	35
1	Los Angeles, CA	35
15	Los Angeles, CA	34
11	New York, NY	34
15	Chicago, IL	33
13	Los Angeles, CA	33
9	Dallas, TX	33
19	Los Angeles, CA	32
2	Miami, FL	32
4	New York, NY	32
18	Chicago, IL	31

23. What Is the Average Demand When Stockouts Occur?
    Insights: On average, the demand for each stockout incident is 300 units.

```sql
SELECT stockout_indicator, ROUND(AVG(actual_demand),2) AS avg_demand
FROM walmart
GROUP BY stockout_indicator;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

stockout_indicator	avg_demand
TRUE	299.44
FALSE	298.71

24. What are the key business performance metrics (total revenue, total quantity sold, total transactions, and average order value)?
    Insights0: There were a total number of 5000 transactions from which the total revenue obtained was ₹15263601.44. With a total number of 14914 items sold, there was an average order value of ₹3052.72. The average order value shows the size of the business operation, whereas the revenue gives an indication of how much is being made through selling goods. The large average order value shows effective pricing and packaging of products, whereas the total quantity shows high demand.

```sql
SELECT SUM(revenue) AS total_revenue, SUM(quantity_sold) AS total_quantity,
COUNT(DISTINCT transaction_id) AS total_transactions, SUM(revenue) * 1.0 / COUNT(DISTINCT transaction_id) AS avg_order_value
FROM walmart;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_revenue	total_quantity	total_transactions	avg_order_value
15263601.449999955	14914	5000	3052.720289999991

25. How do stockouts impact potential sales and which products are most affected?
    Insights: A considerable amount of inventory was out of stock, thereby causing a loss of units and potentially a revenue loss as below for each product category. This shows a definite effect of inadequate inventory levels on business performance. Items that contributed to a larger lost potential revenue show the business areas with huge opportunities for revenue loss.

```sql
SELECT product_name,
    SUM(CASE WHEN stockout_indicator = 'TRUE' THEN quantity_sold ELSE 0 END) AS lost_quantity,
    SUM(CASE WHEN stockout_indicator = 'TRUE' THEN revenue ELSE 0 END) AS potential_lost_revenue
FROM walmart
GROUP BY product_name
ORDER BY potential_lost_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	lost_quantity	potential_lost_revenue
Tablet	1086	1106948.7700000003
Washing Machine	964	1042799.9100000011
Headphones	995	1037042.13
Fridge	1035	1003295.2400000007
TV	950	1002674.0899999997
Smartphone	947	992504.2299999995
Camera	940	977831.1500000004
Laptop	819	844962.1299999994

26. Which products are critical to the business based on both high revenue contribution and frequent stockouts?
Insights: Some of the products generate substantial revenues and are also frequently out of stock. Such products are key to the business because they generate high demand and substantial revenue.

```sql
SELECT product_name, SUM(revenue) AS revenue,
    COUNT(CASE WHEN stockout_indicator = 'TRUE' THEN 1 END) AS stockouts
FROM walmart
GROUP BY product_name
ORDER BY revenue DESC, stockouts DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	revenue	stockouts
TV	2049493.860000001	309
Tablet	1996253.0199999993	352
Fridge	1938012.6900000009	351
Smartphone	1931310.0399999986	330
Washing Machine	1897934.0200000016	316
Camera	1895104.1299999992	318
Headphones	1846334.4499999986	333
Laptop	1709159.2399999984	284

Recommendations:

- Apply demand forecasting for seasonality planning
- Enhance inventory levels of items with higher stockouts
- Develop optimal promotion plan with selective discounting
- Generate diverse income streams among all categories to avoid reliance
- Revamp supply chain management in top-performing zones